

Co1-AT1-Short Answer Test

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Course Code: ITA0501

Course Name: Computer vision

Slot : A

1A) → A suitable technique is image interpolation / image upscaling.

→ Low spatial resolution makes small objects unclear.

Increasing the image size using bilinear or bicubic interpolation estimates missing pixels and makes small objects easier to see.

Suitable Improvement:- Bicubic interpolation gives smoother and clearer results.

2A) → A suitable technique is to use histogram equalization & noise filtering.

→ Histogram equalization improves the brightness and contrast.

→ A median filter removes noise while preserving important edges.

→ This makes objects more visible in the dark image.

Suitable technique:- Histogram equalization with median filtering.

3A) → A suitable technique is to use quantization.

→ Quantization converts continuous intensity values into a fixed number of digital intensity levels.

eg:- An 8-bit image has 256 intensity levels (0-255).

Importance:- It allows the continuous image captured by a camera/sensor to be represented and processed digitally.

4A) → A suitable technique is to use enhancement.

→ Techniques such as adaptive histogram equalization (CLAHE) improve the contrast in different regions of the image.

→ It makes defects more visible even when some parts of the object are brighter (or) darker.

Suitable technique:- CLAHE

5A) → A suitable technique is to use image segmentation.

→ Segmentation divides the image into meaningful regions such as:

Image →

Road	Vehicles	Pedestrians	Background
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→ It helps the autonomous vehicle understand where

the road is and identify objects around it.

Suitable technique:- Semantic Instance Segmentation.

6,A) → A Suitable technique is to use edge detection.

→ The Canny edge detection technique can be applied to identify sudden intensity changes caused by cracks.

Steps:- Image → Grayscale → Noise removal → Canny edge detection → Crack detection.

→ It highlights the boundaries of cracks and helps to identify defective components.

7,A) → A Suitable technique is to use a median filter.

→ A median filter removes noise ~~while~~ while preserving important edges.

→ This is suitable for medical images because tumor boundaries should remain clear after filtering.

8,A) → A Suitable technique is object detection and tracking.

→ Object detection identifies vehicles using techniques such as YOLO. Tracking then follows each vehicle.

Across consecutive frames.

Process:-

Video $\rightarrow$ vehicle detection $\rightarrow$ vehicle tracking $\rightarrow$ Count/monitor vehicles

This helps in traffic monitoring, congestion detection and traffic management.

9A) $\rightarrow$ The possible cause is motion blur.

$\rightarrow$ When an object moves during image capture, its position changes while the camera shutter is open, producing a blurred image.

Suitable technique:- Image deblurring/restoration, such as inverse filtering or wiener filtering.

$\rightarrow$ It attempts to restore the sharpness of the moving object.

10A) $\rightarrow$ A suitable technique is to use image classification.

$\rightarrow$ The captured product image is processed and important features are extracted. A classifier then categorizes it as:

Product image $\rightarrow$ preprocessing $\rightarrow$ feature extraction $\rightarrow$ classification $\rightarrow$ Good/Defective.

$\rightarrow$ Machine Learning classifiers such as SVM, Decision-

Tree (or) CNN can be used.

→ This supports automatic quality inspection, reduces human error and allows faster inspection.